

A two-replica proof of the Krenn–Gu conjecture

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Abstract

In the graph model of photonic state preparation, the amplitude of a vertex colouring is a sum of products of complex edge weights over perfect matchings. The Krenn–Gu conjecture asks whether these weights can cancel every mixed colouring while retaining at least three monochromatic colourings on more than four vertices. We prove that they cannot. The main difficulty is that the existence of a mixed matching alone does not prevent its amplitude from cancelling. Our argument forces a stronger structure: after parallel edges with the same endpoint colours are aggregated, every edge is monochromatic and every active colour forms a perfect matching. A vertex colouring then has at most one contributing matching, so a known combinatorial obstruction finishes the proof. The structural argument uses two identical formal copies of the matching polynomial. Their symmetries impose polynomial identities that force each supported edge to contribute the entire amplitude of its colour. Complex weights, parallel edges, and distinct colours at the two ends of an edge are all allowed. A Lean formalization verifies the normalized equation-system statement at every even order at least six and every palette size at least three.

1 Introduction

Perfect matchings provide a bridge between graph theory and the preparation of entangled states with photon-pair sources. Vertices represent output paths, edges represent pair contributions, and endpoint colours specify the modes. Selecting one photon at each output corresponds to selecting a perfect matching. Contributions that induce the same output modes add as complex amplitudes; they can therefore interfere destructively. The resulting existence question was formulated by Krenn, Gu and Soltész [1, Question 1]; see also the matching examples in [4, Section 2].

For a GHZ target, the desired colourings assign the same colour to every vertex. All other colourings must have amplitude zero. Graph theory can force an unwanted matching to exist, but complex weights might cancel its contribution against other matchings. The proof must therefore explain why such cancellation cannot satisfy all the target equations simultaneously. We do this by deriving a structural restriction on the weighted graph itself.

1.1 Matching amplitudes and the conjecture

Let $G = (V, E)$ be a finite loopless multigraph. Each edge $e = uv$ has weight $w(e) \in \mathbb{C}$ and two endpoint colours $\chi_u(e), \chi_v(e)$. We write $[d] = \{1, \dots, d\}$ for the target palette and $\mathcal{M}(G)$ for the set of perfect matchings of G . For $M \in \mathcal{M}(G)$, let $e_M(v)$ be its unique edge at v . The matching

induces the vertex colouring $v \mapsto \chi_v(e_M(v))$. For a vertex colouring $\kappa : V \rightarrow [d]$, define

$$W(\kappa) = \sum_{\substack{M \in \mathcal{M}(G) \\ \chi_v(e_M(v)) = \kappa(v) \text{ for every } v}} \prod_{e \in M} w(e).$$

An empty sum is zero. We write i^V for the colouring that is constantly i . A colouring that is not constant is called mixed. For a matrix B and a vertex set S , $B[S]$ denotes its principal submatrix on S .

Definition 1.1 (Generalized GHZ condition). The weighted half-edge-coloured graph satisfies the generalized GHZ condition on $[d]$ if $W(i^V) = \tau_i \neq 0$ for every $i \in [d]$, and $W(\kappa) = 0$ for every mixed vertex colouring $\kappa : V \rightarrow [d]$.

The usual normalized condition requires $\tau_i = 1$. Our definitions follow Chandran, Gajjala and Illickan [2, Sections 1.1 and 1.4]; the generalized condition permits unequal nonzero amplitudes.

Theorem 1.2 (Krenn–Gu conjecture). *If $|V| > 4$ is even, there is no complex weighted half-edge-coloured multigraph satisfying the generalized GHZ condition on a palette of size $d \geq 3$.*

The theorem concerns this matching model with arbitrary complex weights. There is no restriction on edge density or on the number of parallel edges. It suffices to retain three active colours: removing edges with other endpoint colours leaves every amplitude on the retained palette unchanged.

1.2 Cancellation and the four-vertex boundary

Consider the left-hand graph in Figure 1. Both of its perfect matchings induce the mixed colouring (a, a, b, b) . Their weights are $1 \cdot 1$ and $1 \cdot (-1)$, so

$$W(a, a, b, b) = 1 \cdot 1 + 1 \cdot (-1) = 0.$$

Thus a zero amplitude does not imply the absence of a matching. This small graph illustrates cancellation only; it does not satisfy the GHZ condition. The proof has to use the nonzero monochromatic amplitudes together with the vanishing of *every* mixed amplitude.

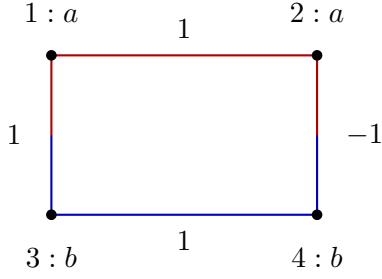
The restriction to more than four vertices is necessary. In Figure 1(b), the three perfect matchings of K_4 are

$$\{12, 34\}, \quad \{13, 24\}, \quad \{14, 23\}.$$

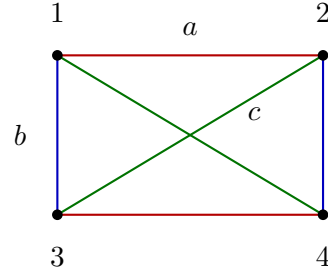
Colour these matchings a, b, c , respectively, and give every edge weight one. The only induced colourings are a^V, b^V, c^V , each with amplitude one. Our structural argument will recover precisely the feature visible in this example: each colour forms a perfect matching. The final graph obstruction then distinguishes four vertices from larger orders.

1.3 Related work

Chandran and Gajjala use edge pruning and a sparse representative subgraph to bound the dimension of simple GHZ experiment graphs [3]. Chandran, Gajjala and Illickan prove the conjecture for graphs of vertex connectivity at most two and for graphs of maximum degree at most three. Their reduction also implies that a counterexample with the fewest vertices must be four-connected [2, Theorems 9–11]. These results concern the same weighted model.



(a) Two nonzero matching contributions cancel.



(b) The four-vertex GHZ construction.

Figure 1: The obstruction and the exceptional case. In (a), labels beside edges are weights and labels at vertices give the inherited colours; the two vertical edges have different colours at their ends. In (b), all weights are one. Horizontal, vertical, and diagonal edges have colours a, b, c , respectively. The diagonal crossing is not a vertex.

A separate result, attributed to Bogdanov in [2, Theorem 7], states that three monochromatic perfect matchings of distinct colours on more than four vertices force a mixed perfect matching. This is an existence statement. In the complex weighted setting, different matchings may cancel, so it does not by itself settle the conjecture. Our structural argument removes this possibility before invoking the combinatorial obstruction.

1.4 Proof overview: forcing one edge per colour

We first remove an ambiguity caused by parallel edges. For fixed vertices u, v and fixed endpoint colours i, j , replace all corresponding parallel edges by one edge whose weight is their sum. Delete that edge if the sum is zero. A perfect matching uses at most one edge between u and v , so distributivity shows that this operation preserves every $W(\kappa)$. All support and uniqueness statements below refer to this *aggregate graph*.

The argument has two structural steps and a short combinatorial conclusion. We describe the conclusion first far enough to make the role of each structural step explicit.

Step 1: every remaining edge has equal endpoint colours. Lemma 4.2 proves that every aggregate weight with distinct endpoint colours is zero. After this reduction, fix an active colour and write B_{pq} for its weight on the edge pq . A supported edge means one with $B_{pq} \neq 0$. Let $\beta \neq 0$ be the amplitude of the monochromatic colouring in this colour.

Step 2: each supported edge contributes the entire amplitude. For an edge pq , define its matching subtotal

$$C_{pq} = \sum_{\substack{M \text{ a perfect matching of this colour} \\ pq \in M}} \prod_{e \in M} w(e).$$

The central identity, proved in Theorem 5.1, is

$$\boxed{C_{pq} = \beta \quad \text{whenever } B_{pq} \neq 0.}$$

Its consequence is immediate. Every perfect matching uses exactly one edge at a fixed vertex p . If $\deg_B(p)$ counts the supported edges of this colour at p , then

$$\beta = \sum_{q: B_{pq} \neq 0} C_{pq} = \deg_B(p) \beta.$$

Since $\beta \neq 0$, we obtain $\deg_B(p) = 1$. This applies to every vertex and every active colour: each colour graph is a perfect matching. In particular, the conclusion uses complex amplitudes directly and requires no positivity argument.

Step 3: a mixed matching can no longer cancel. Given a vertex colouring κ , its colour at each vertex specifies the only possible incident edge in a contributing matching. Consequently there is at most one such matching. If it exists, its weight is a product of nonzero edge weights and hence is nonzero. But three monochromatic perfect matchings of distinct colours on more than four vertices force a mixed perfect matching [2, Theorem 7]. Its nonzero amplitude contradicts the GHZ condition. We include the graph argument in Section 7.

Where the algebra enters. The difficult work is to establish Steps 1 and 2. A matching subtotal can be written as

$$C_{pq} = B_{pq} \text{haf}(B[V \setminus \{p, q\}]),$$

where the hafnian is the sum of products over perfect matchings on the remaining vertices. Thus Step 2 is an identity relating a full matching sum to a sum with two vertices removed. The remaining graph is not assumed to satisfy any GHZ condition; the identity must follow from the equations on the original graph.

We obtain such identities by taking two identical formal copies of the matching polynomial. Mixing the copies preserves their pairing data, while certain alternating expressions change sign. This yields vanishing identities that are not visible in a single matching sum. A second use of the symmetry restricts an auxiliary matrix to a small family of polynomial forms. The original matching equations then impose differential equations on their coefficients. A polynomial of positive degree cannot satisfy $p' + \delta p = k$ when $\delta \neq 0$ and k is constant: its highest-degree term would have nothing to cancel. This finite-degree argument gives the supported-edge identity.

Section 2 develops the pairing language from a four-factor example. Sections 3 and 4 establish Step 1; Section 5 proves the identity in Step 2. The final two steps above are completed in Sections 6 and 7. Section 8 states the precise formalization scope.

2 Algebraic formulation and formal pairings

2.1 Encoding a matching sum

Our first algebraic operation enforces the matching rule: a vertex may be used only once. Associate a variable $x_{v,i}$ with colour i at vertex v , and set every product $x_{v,i}x_{v,j}$ to zero. Thus a product of edges sharing a vertex vanishes, regardless of their colours. Products of vertex-disjoint edges survive.

Write $n = |V|$ and fix an ordering of V . Let $A_{uv}(i, j)$ be the sum of the weights of edges whose endpoint colours are i at u and j at v , as in the aggregation in Section 1.4. The algebra and its

edge polynomial are

$$\begin{aligned} \mathcal{A}_V &= \mathbb{C}[x_{v,i} : v \in V, i \in \{a, b, c\}] / (x_{v,i}x_{v,j} : v \in V, i, j \in \{a, b, c\}), \\ A &= \sum_{u < v} \sum_{i, j \in \{a, b, c\}} A_{uv}(i, j) x_{u,i} x_{v,j}. \end{aligned}$$

Every surviving term in its divided top power chooses each vertex exactly once. The division by $(n/2)!$ removes the ordering of the chosen edges. Thus the coefficient of $\prod_v x_{v, \kappa(v)}$ is exactly $W(\kappa)$. We use $A^{[j]} = A^j/j!$ for these divided powers. For one colour on four vertices, the coefficient is the familiar three-term matching sum

$$\text{haf}(B) = B_{12}B_{34} + B_{13}B_{24} + B_{14}B_{23}.$$

In general, $\text{haf}(B[S])$ denotes the corresponding sum on the even vertex set S , with value one when S is empty.

We may therefore express the ternary instance of Definition 1.1 as follows. Write a_v, b_v, c_v for the three coordinates at v , and $h^S = \prod_{v \in S} h_v$.

Theorem 2.1 (Algebraic form of Theorem 1.2). *Let V have even cardinality $n > 4$. In the algebra $\mathcal{A}_V$, let A be any quadratic involving only distinct vertices, with arbitrary complex coefficients. Then*

$$A^{[n/2]} = \tau_a a^V + \tau_b b^V + \tau_c c^V, \quad \tau_a \tau_b \tau_c \neq 0 \quad (1)$$

is impossible.

The coefficient interpretation just proved identifies this statement with the graph theorem after palette restriction. The algebra keeps the original complex weights, including all aggregate bichromatic weights.

2.2 Normalizing nonzero target amplitudes

The written argument works directly with the nonzero τ_i in (1). To compare it with the formally checked statement, it is useful to give the normalization explicitly. Fix one vertex r . Multiply each edge incident on r , with colour i at that endpoint, by τ_i^{-1} . Every perfect matching contains exactly one such edge, so the new amplitude of a colouring κ is

$$W'(\kappa) = \tau_{\kappa(r)}^{-1} W(\kappa).$$

Every pure amplitude becomes one and every mixed amplitude remains zero. Conversely, a normalized solution already satisfies the generalized condition. This proves the equivalence without choosing roots of complex numbers; compare the scaling lemma in [2, Lemma 14].

2.3 Formal pairings: a four-factor example

A second algebraic description lets us exploit symmetry. Assign symmetric pairing coefficients $c_{ij} = c_{ji}$ to formal symbols z_i, z_j . Define the pairing sum, also called a formal Wick moment, by summing over all pairings of the labelled occurrences. For four factors it is

$$\mathcal{W}(z_1 z_2 z_3 z_4) = c_{12} c_{34} + c_{13} c_{24} + c_{14} c_{23}.$$

This is exactly a matching sum. For any even number of factors the same rule applies; an odd number has moment zero, and the empty product has moment one. Repeated symbols still have distinct labelled occurrences. We extend the rule multilinearly to linear combinations of symbols.

The coefficients c_{ij} are traditionally called covariances, but here they are arbitrary complex numbers or polynomial parameters. No probability distribution, positivity, or invertibility is required. This is the algebraic pairing formula underlying Isserlis's moment formula [5]. In particular, a linear substitution preserving all pairwise covariances preserves every moment: each summand uses only those covariances.

2.4 Why take two copies?

Take two copies X_i, Y_i with identical within-copy covariances and zero cross-copy covariances. For example, the change of variables

$$X'_i = \frac{X_i + Y_i}{\sqrt{2}}, \quad Y'_i = \frac{X_i - Y_i}{\sqrt{2}}$$

preserves all pairing data: both new copies have the original covariances, and their cross covariances are still zero. More generally, any complex orthogonal matrix acting on the two copy labels has this property. The graph vertices and edge weights remain fixed.

There is also an expression that detects the determinant of this change of copy coordinates. Choose distinct colours α_i, β_i at a vertex and form

$$\Delta_i = X_{i,\alpha_i} Y_{i,\beta_i} - X_{i,\beta_i} Y_{i,\alpha_i}.$$

Under a copy transformation O , this expression becomes $(\det O)\Delta_i$. A reflection therefore changes the sign of a product of Δ_i 's on an odd set of vertices. If the remaining factors in a moment are fixed by that reflection, invariance of the moment forces it to vanish. Section 3 chooses those factors and the reflection to obtain identities for the original source rows.

On an even set of vertices the product instead stays unchanged. This gives the invariant pairing used in Section 5. Its polynomial dependence on auxiliary parameters is essential: degree comparisons turn the symmetry constraints into exact identities. All parameter variations take place in these auxiliary expressions, with the original edge weights held fixed. We never assume that deleting vertices preserves the GHZ condition.

3 Whole binary higher responses vanish

The following lemma combines the reflection and common-factor argument of [7], the two-colour mixed-coefficient identity of [8], and the finite-degree pure-response argument of [9]. We give the complete proof, including the terminal response.

Lemma 3.1 (Whole binary higher-response vanishing). *Let Ω have $2m + 1 \geq 3$ vertices, let R be an arbitrary quadratic on Ω , and let D be a vector space of rows L such that*

$$\Phi(L) = LR^{[m]} = \sum_h f_h(L) h^\Omega. \quad (2)$$

Assume all three linear functions f_h are nonzero and at least two are independent. For odd $p \leq 2m + 1$, define

$$H_p(L) = L^p R^{[(2m+1-p)/2]}.$$

For every $p \geq 3$, the tensor $H_p(L)$ has zero coefficient on every word using at most two colours, including the terminal degree $p = 2m + 1$. The same vanishing holds after polarization for products of arbitrary rows from the same D .

Proof. Give formal variables $X_{i,h}$ their R covariances, zero covariances within a vertex, and give an auxiliary g covariance $L_i(h)$ with $X_{i,h}$ and zero covariance with itself. A Wick moment with p copies of g and one specified coordinate at each vertex is the corresponding coefficient of $H_p(L)$. The $p!$ assignments of auxiliary occurrences are precisely those of the raw power L^p ; the divided power of R counts each remaining matching once.

Take two independent identical replicas (X, g_1) , (Y, g_2) . Choose distinct local colours α_i, β_i at each vertex and set

$$\Delta_i = X_{i,\alpha_i} Y_{i,\beta_i} - X_{i,\beta_i} Y_{i,\alpha_i}.$$

Over $\mathbb{C}(s, t)$, the orthogonal reflection

$$\frac{1}{s^2 + t^2} \begin{pmatrix} s^2 - t^2 & 2st \\ 2st & t^2 - s^2 \end{pmatrix}$$

fixes $sg_1 + tg_2$ and negates each Δ_i . It preserves the replicated covariance, so the moment of

$$(sg_1 + tg_2)^{p+1} \prod_{i \in \Omega} \Delta_i$$

is zero: there are an odd number of determinants. Extracting $s^p t$ gives

$$\sum_{I \subseteq \Omega} (-1)^{|I|} H_p(L)(w_I) \Phi(L)(w'_I) = 0, \quad (3)$$

where w_I uses β on I and α elsewhere, and w'_I is complementary. Equality over $\mathbb{C}(s, t)$ gives a polynomial identity, including isotropic parameter values by extension. No operation on the original source is being made.

With constant local pair a, b , (3) says

$$f_b[a^\Omega] H_p = f_a[b^\Omega] H_p.$$

Two independent f 's are coprime in $\mathbb{C}[D]$, hence for $p = 2r + 1$ there is a common homogeneous polynomial q_r of degree $2r$ such that $[h^\Omega] H_p = f_h q_r$ for every h ; $q_0 = 1$. For a mixed word w avoiding an active colour h , choose $\alpha_i = w_i$, $\beta_i = h$. The only constant complementary word in (3) is h^Ω , giving $f_h[w] H_p = 0$. Polynomial cancellation proves all binary mixed coefficients zero, even at individual rows where $f_h(L) = 0$.

Define finite polynomials

$$\Psi(L) = \sum_{r=0}^m \frac{H_{2r+1}(L)}{(2r+1)!}, \quad g(L) = \sum_{r=0}^m \frac{q_r(L)}{(2r+1)!}.$$

On any binary palette a, b , $\Psi(L)$ thus equals

$$g(L)(f_a(L)a^\Omega + f_b(L)b^\Omega).$$

For two tensors on Ω , let K be the product of local alternating pairings, normalized by $K(a^\Omega, b^\Omega) = 1$. Wick moments with shifted variables $X_i + L_i$ show that $K(\Psi(L), \Psi(M))$ is invariant under

simultaneous $\text{SO}(2)$ rotation of the two mean rows: the replica covariance and each determinant are unchanged. Choose independent f_a, f_b . This pairing is

$$(f_a(L)f_b(M) - f_b(L)f_a(M))g(L)g(M).$$

The determinant factor is a nonzero polynomial and is rotation invariant. Cancel it in $\mathbb{C}[D \times D]$, then rotate by 45 degrees and set $M = 0$. Since g is even and $g(0) = 1$,

$$g(L/\sqrt{2})^2 = g(L).$$

If g had positive degree, the degrees of the two sides would differ. Thus $g = 1$, and all q_r with $r \geq 1$ vanish. Together with (3), this proves whole binary vanishing through the terminal degree $2m + 1$. Polarization gives the same statement for products of arbitrary rows from this same D . $\square$

For a source (1), delete any root p . Its three actual incident rows have responses $\tau_h h^{V \setminus \{p\}}$, so their span satisfies (2) with three independent f_h . Thus Lemma 3.1 applies at every original root, before diagonalization.

4 Even omission identities and diagonal reduction

The even-omission argument originates in [10]. Its role is to convert the rooted identities into off-diagonal matrix identities.

Lemma 4.1 (Even omission). *In the setting of (2), omit a vertex q , put $S = \Omega \setminus \{q\}$, and write*

$$\begin{aligned} R &= Q + \sum_h h_q V_h, & L &= U + \sum_h d_h h_q, \\ E(U) &= \sum_{r=0}^m \frac{U^{2r}}{(2r)!} Q^{[m-r]}, & F &= E(0). \end{aligned}$$

For any $L_1, L_2 \in D$ and any active colour k ,

$$[k^S] U(L_1) U(L_2) Q^{[m-1]} = 0.$$

This includes $m = 1$.

Proof. The full coefficient of $q = h$ in $\Psi(L)$ is

$$d_h E(U) + E_{V_h}(U),$$

where subscripts mean directional derivatives in the mean, with Q fixed. By Lemma 3.1, its projection onto any palette h, k on S is $f_h(L)h^S$. All the following pairings use this binary projection implicitly.

Because $|S|$ is even, K is symmetric and $K(Z, h^S) = [k^S]Z$. The same two-replica rotation gives

$$\begin{aligned} K(E(U), E(U)) &= K(F, E(\sqrt{2}U)), \\ K(E(U), E_V(U)) &= \frac{1}{\sqrt{2}} K(F, E_V(\sqrt{2}U)). \end{aligned} \tag{4}$$

The second identity follows by differentiating the rotation identity at means $U, U + tV$: the second rotated mean is zero at $t = 0$ and $E'(0) = 0$. Substitute

$$E_{V_h}(U) = f_h h^S - d_h E(U)$$

and the corresponding identity for $\sqrt{2}L$ into the second equation of (4). The first cancels the d_h terms, leaving

$$f_h(L)[k^S](E(U) - F) = 0.$$

Cancel the nonzero polynomial f_h . Separating homogeneous degrees and polarizing yields

$$[k^S] U(L_1)U(L_2)Q^{[m-1]} = 0. \quad (5)$$

□

We next apply this identity to the actual hafnian cofactors. This is the diagonal reduction proved in [11].

Lemma 4.2 (Global diagonal reduction). *Every endpoint-colour edge block of a source (1) is diagonal in the given target coordinates.*

Proof. For a fixed colour h , define

$$\begin{aligned} M_h[p, r] &= A_{pr}(h, h) & (p \neq r), & \quad M_h[p, p] = 0, \\ C_h[r, q] &= \text{haf}(M_h[V \setminus \{r, q\}]) & (r \neq q), & \quad C_h[q, q] = 0, \\ B_{ih}[p, r] &= A_{pr}(i, h) & (p \neq r), & \quad B_{ih}[p, p] = 0. \end{aligned}$$

Expansion of the original one-defect word at p gives

$$(B_{ih}C_h)[p, p] = \delta_{i,h}\tau_h.$$

For $p \neq q$, expand each $C_h[r, q]$ at its retained vertex p :

$$\begin{aligned} (B_{ih}C_h)[p, q] &= \sum_{\substack{r, s \in V \setminus \{p, q\} \\ r \neq s}} A_{pr}(i, h) A_{ps}(h, h) \text{haf}(M_h[V \setminus \{p, q, r, s\}]) \\ &= 0. \end{aligned} \quad (6)$$

The last expression is exactly (5) for two rows of the original p family after omitting q . Both sums are ordered; there is no factor $1/2$. Consequently

$$B_{ih}C_h = \delta_{i,h}\tau_h I.$$

Taking $i = h$ proves C_h invertible. Taking $i \neq h$ then proves $B_{ih} = 0$. Every original edge block is therefore diagonal. This uses neither cofactor nonvanishing assumptions nor a change of basis. □

5 The supported-edge identity

The construction in this section is the finite two-replica endpoint argument of [12]. Its inputs are the original root identities obtained above; the auxiliary parameters do not change the edge weights.

Fix two colours B, H after diagonal reduction and roots $p \neq q$. Put $U = V \setminus \{p, q\}$, $N = |U|$ (even), and let Q be the literal binary quadratic on U . Write

$$\begin{aligned} d &= B_{pq}, & e &= H_{pq}, & \beta &= \text{haf}(B), \\ \eta &= \text{haf}(H), & \alpha &= \text{haf}(B[U]), \\ x &= \sum_{i \in U} B_{pi} b_i, & y &= \sum_{i \in U} B_{qi} b_i, \\ u &= \sum_{i \in U} H_{pi} h_i, & v &= \sum_{i \in U} H_{qi} h_i, \\ E(L) &= \sum_{j=0}^{N/2} \frac{L^{2j}}{(2j)!} Q^{[N/2-j]}, & F &= E(0). \end{aligned}$$

Theorem 5.1 (Supported endpoint identity). *For every supported edge $B_{pq} \neq 0$,*

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

Derivatives of E always keep Q fixed. Lemma 3.1, applied to the original p and q row planes and expanded at the other root, gives the exact polynomial identities

$$\begin{aligned} E_y(sx + tu) + dsE(sx + tu) &= s\beta b^U, \\ E_v(sx + tu) + etE(sx + tu) &= t\eta h^U, \\ E_x(ay + bv) + daE(ay + bv) &= a\beta b^U, \\ E_u(ay + bv) + ebE(ay + bv) &= b\eta h^U. \end{aligned} \tag{7}$$

These include the highest coefficient, which uses the terminal original odd response of degree $N + 1$. They are identities in mean parameters on the unchanged source. Differentiating those parameters is legitimate and gives

$$\begin{aligned} E_u(ay) &= 0, & E_v(sx) &= 0, \\ E_{yu}(sx) &= -dsE_u(sx), \\ E_{uv}(sx) + eE(sx) &= \eta h^U, \\ E_{xy}(0) + dF &= \beta b^U. \end{aligned} \tag{8}$$

On tensors on U , use the symmetric binary alternating pairing K , normalized by $K(b^U, h^U) = 1$. The polynomial $K(E(L_1), E(L_2))$ is invariant under the full $O(2)$ action on the two means. The Wick proof is as in Section 3; a determinant -1 transformation contributes $(-1)^N = 1$.

Introduce independent columns $P, Q_c, R_c, S_c \in \mathbb{C}^2$ and means $L_i = P_i x + (Q_c)_i y + (R_c)_i u + (S_c)_i v$. Set

$$\begin{aligned} G &= K(E(L_1), E(L_2)), & f &= G|_{R_c=S_c=0}, \\ M_{ij} &= \partial_{(R_c)_i} \partial_{(S_c)_j} G|_{R_c=S_c=0}, & T &= M + efI. \end{aligned}$$

The entry M_{ij} records an insertion of the H -row u in copy i and of the H -row v in copy j . When both insertions belong to the same copy, they represent the two roots being matched into U . There is one more possibility: the roots are matched directly to each other by the H -edge of weight e , leaving the retained polynomial unchanged. That contribution is ef on each diagonal entry. Thus the correction in T combines the two terms of the original matching expansion, just as $E_{uv} + eE$ does in (8). There is no direct edge between distinct copies.

These are finite polynomials. Orthogonal invariance gives

$$M(OP, OQ_c) = OM(P, Q_c)O^T,$$

and similarly for T . By (8),

$$M_{12} = K(E_u(P_1x + (Q_c)_1y), E_v(P_2x + (Q_c)_2y))$$

is divisible by $P_1(Q_c)_2$; likewise M_{21} is divisible by $P_2(Q_c)_1$.

Lemma 5.2 (Polynomial orthogonal kernel normal form). *For the kernel matrix just defined,*

$$\begin{aligned} T &= aI + bPQ_c^T, \\ a, b &\in \mathbb{C}[\sigma, \tau, c], \\ \sigma &= P \cdot P, \quad \tau = Q_c \cdot Q_c, \quad c = P \cdot Q_c. \end{aligned} \tag{9}$$

Proof. In an orthonormal frame where $P_1 = 0$, $M_{12} = 0$, so MP is parallel to P . In a frame where $(Q_c)_2 = 0$, $M_{12} = 0$, so $M^T Q_c$ is parallel to Q_c . Such frames exist on a dense open set of nonisotropic columns. Their eigenvalues agree when $P \cdot Q_c \neq 0$. For two independent columns in dimension two, these conditions force

$$M = a_0I + bPQ_c^T.$$

Indeed, M minus the common eigenvalue times I annihilates P and has left kernel Q_c ; its rank-one form is a multiple of $(P \cdot Q_c)I - PQ_c^T$. Literal divisibility proves that

$$b = \frac{M_{12}}{P_1(Q_c)_2}, \quad a_0 = M_{11} - bP_1(Q_c)_1$$

are polynomials. The generic identity therefore extends everywhere. Uniqueness and covariance make a_0, b scalar $O(2)$ invariants, and the same is true of $a = a_0 + ef$.

The scalar invariant-ring assertion is the two-vector case of the first fundamental theorem for the orthogonal group; see [6, Section 10.2(a)]. For completeness, we prove this particular case directly. In coordinates $p_+ = P_1 + iP_2$, $p_- = P_1 - iP_2$ and q_+, q_- , $SO(2)$ acts with weights $+1, -1$. Its invariant monomials are generated by $\sigma, \tau, r = p_+q_-, s = p_-q_+$, with $rs = \sigma\tau$. Reflection exchanges r, s . Symmetric polynomials reduce to $r + s = 2c$ and $rs = \sigma\tau$, giving $\mathbb{C}[\sigma, \tau, c]$. These three Gram coordinates are algebraically independent. $\square$

Lemma 5.3 (Polynomial differential equations). *Let R be a commutative integral domain, let $d \in R$ be nonzero, and let $p \in R[z]$. If $p' + dp = 0$, then $p = 0$. If $p' + dp = k$ for $k \in R$, then p is constant and $dp(0) = k$.*

Proof. If $p \neq 0$ has degree r , the coefficient of z^r in $p' + dp$ is d times the leading coefficient of p , which is nonzero. This excludes a homogeneous equation. In the inhomogeneous equation it excludes $r > 0$; substituting a constant polynomial gives $dp(0) = k$. $\square$

The coefficient domain may itself be a polynomial ring. This elementary step, including its endpoint cancellation consequence, is formalized in *PolynomialODE.lean* [14].

Proof of Theorem 5.1. At $(Q_c)_1 = 0$, differentiate M_{12} in $(Q_c)_1$ and use (8). One obtains

$$\partial_{(Q_c)_1} M_{12} \Big|_{(Q_c)_1=0} = -dP_1 M_{12} \Big|_{(Q_c)_1=0}, \quad P_1^2(Q_c)_2(b_c + db) = 0.$$

The substitution

$$(\sigma, \tau, c) = (P_1^2 + P_2^2, (Q_c)_2^2, P_2(Q_c)_2)$$

is injective on polynomial rings: every $\tau \neq 0$ and arbitrary σ, c is attained over $\mathbb{C}$. Thus $b_c + db = 0$. Apply Lemma 5.3 over $\mathbb{C}[\sigma, \tau]$, with polynomial variable c . Since $d \neq 0$, it gives $b = 0$, and hence $T = aI$.

Write

$$Z = E_{uv}(P_2x + (Q_c)_2y) + eE(P_2x + (Q_c)_2y).$$

The full expression is $T_{22} = K(E(P_1x + (Q_c)_1y), Z)$. Differentiate this expression in $(Q_c)_1$, then set $(Q_c)_1 = 0$. Equation (7) gives

$$\partial_{(Q_c)_1} T_{22} \Big|_{(Q_c)_1=0} = P_1 \beta K(b^U, Z) - dP_1 T_{22} \Big|_{(Q_c)_1=0}.$$

The pure- H coefficient of Z is η : positive B mean insertions cannot contribute to an all- H word, so it is the same as at zero, where (8) gives $E_{uv}(0) + eF = \eta h^U$. Thus $K(b^U, Z) = \eta$. Using (9) and the same injective substitution proves

$$a_c + da = \beta \eta.$$

Lemma 5.3, again over $\mathbb{C}[\sigma, \tau]$, gives $a = \beta \eta / d$, independent of all three Gram variables. At the origin,

$$a(0) = K(F, E_{uv}(0) + eF) = \eta K(F, h^U) = \eta \alpha.$$

Since $\eta \neq 0$, this proves

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]) \quad \text{whenever } B_{pq} \neq 0. \quad (10)$$

Throughout the argument the edge weights are fixed. All differentiations are in the auxiliary columns P, Q_c , and the polynomial identities are finite. $\square$

6 Every colour is a perfect matching

Corollary 6.1 (Matching rigidity of every active colour). *Each colour graph of a source (1) is a weighted perfect matching.*

Proof. The ordinary hafnian expansion at a vertex p says

$$\beta = \sum_{q \neq p} B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

By (10), every nonzero summand equals β . Since $\beta \neq 0$, the number of supported B neighbors of p is exactly one. This holds at every vertex, and for each of the three colours by choosing any other active colour as H . $\square$

Two such matchings cannot share an edge pq : choosing that edge in colour B and the other matching in colour H off p, q gives a mixed word with nonzero coefficient, the product of nonzero matching weights. Consequently their union is a simple cubic graph properly coloured with three colours. Any receiving colour word supports at most one matching, because each vertex has exactly one incident edge of its requested colour. Therefore a mixed perfect matching would have a nonzero coefficient and violate (1).

7 The remaining graph lemma

The following lemma is a special case of the obstruction attributed to Bogdanov in [2, Theorem 7]. At this point, each colour is already a perfect matching, and no vertex colouring can receive cancelling matching contributions. We include a proof for this special case.

Lemma 7.1 (Three-matching obstruction). *Three pairwise disjoint perfect matchings F, G, H on $n > 4$ vertices have a mixed perfect matching in their union.*

Proof. Suppose otherwise. The union $F \cup G$ must be a single alternating Hamilton cycle C : otherwise choosing F on one component and G on the others already gives a mixed matching. Label its vertices $0, \dots, n-1$ cyclically. The matching H consists of chords.

A chord with endpoints of opposite parity leaves two even paths when its endpoints are deleted. Matching those paths along C and adding the chord gives a mixed perfect matching. Hence all H chords join equal parities. The two parity classes must each have even size; otherwise H is impossible.

An even-even chord and an odd-odd chord whose endpoints interlace leave four even paths on deleting their endpoints. Those paths and the two chords give a mixed perfect matching when $n > 4$. Thus no such pair interlaces.

Choose a chord and one of its sides with the fewest vertices strictly inside, over all chords and both sides. Its endpoints have equal parity, so its interior contains a vertex of the opposite parity. Every vertex of that opposite parity in the interior must be H -matched to another interior vertex; a partner outside would give an interlacing opposite-parity chord. Therefore some H chord lies strictly inside the chosen arc. One side of that chord has fewer interior vertices, a contradiction. $\square$

This supplies a mixed perfect matching, contradicting Section 6. Equation (1) is impossible for every even $n > 4$, completing the proof of Theorem 2.1. The graph interpretation in Section 2 and restriction to three active colours now give Theorem 1.2.

8 Boundary cases and formal verification

For $n = 4$, the three matchings of K_4 give the permitted ternary source. In Section 7, a pair of interlacing chords then exhausts the vertices and gives the pure third matching, so no contradiction is asserted. Sections 3–6 remain valid at $n = 4$, including their terminal response coefficients. For $n = 2$, the odd-core lemma does not apply and arbitrary target dimension is possible. Odd n has no perfect matchings in this model.

The historical sources [7, 8, 9, 10, 11, 12] are internal research notes from this repository. Their proofs have been included here, so these citations identify provenance rather than replace mathematical steps. The published external inputs are the graph formulation and combinatorial obstruction [1, 2]; the pairing and invariant-theory references [5, 6] provide background for identities also proved in the text.

The Lean development in [14] proves nonexistence for every even $N \geq 6$ and every $D \geq 3$, with arbitrary complex aggregate endpoint-colour weights and the normalized equations

$$W(\kappa) = \begin{cases} 1, & \kappa \text{ is constant,} \\ 0, & \kappa \text{ is mixed.} \end{cases}$$

The final declaration is `UpstreamFullProof.eqSystem_no_solution_ge6_ge3` in `FullProof.lean`, within the namespace `KrennAllOrders`. An adapter identifies the local matching sums with the exact `MonochromaticQuantumGraph.WeightsN` and `EqSystemN` definitions in the pinned Formal Conjectures source. The final theorem has no remaining source-reduction or structural hypothesis.

The checked proof covers the complete chain: the matching model, copy symmetries, response and omission identities, diagonal reduction, the polynomial kernel argument, the supported-edge identity, and the graph obstruction. The real-to-complex transfer and the corresponding real-weight corollaries are also formalized.

The dependency audit reports only three standard axioms: `propext`, `Classical.choice`, and `Quot.sound`. There are no proof holes or additional axioms in the final dependencies.

The final Lean statement uses aggregate weights and unit pure amplitudes. Sections 1.4 and 2.2 give the explicit mathematical translations from the multigraph formulation and arbitrary nonzero pure amplitudes used here. These explanatory translations should be distinguished from the exact interface of the linked Lean theorem. The artifact includes the pinned toolchain, source hashes, axiom reports, and commands for reproducing both the local proof and the upstream adapter.

The written argument is based on the frozen mathematical source in [13]. The associated agent audits are internal research records; they do not constitute external peer review. This manuscript develops its exposition independently of those historical records.

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